

Literature-Implied Negative Answer to a Triple Rademacher Tensor Formula in arXiv:1103.2090

Run fa_banach_001, lane 19

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Status

This packet records a literature-implied negative subcase for a question in Pisier, arXiv:1103.2090, *Random Series of Trace Class Operators*. The negative subcase follows from the later theorem of Briet–Naor–Regev, arXiv:1208.0539, together with the Maurey–Pisier cotype criterion recalled in the source paper.

Source Question

In the final remarks of arXiv:1103.2090, after the two-fold formula

$$R(E \widehat{\otimes} F) \approx R(E) \widehat{\otimes} F + E \widehat{\otimes} R(F),$$

Pisier asks whether there is a similar result for $R(E \widehat{\otimes} F \widehat{\otimes} G)$. The displayed guess is

$$R(E) \widehat{\otimes} F \widehat{\otimes} G + E \widehat{\otimes} R(F) \widehat{\otimes} G + E \widehat{\otimes} F \widehat{\otimes} R(G).$$

Supporting Result

Briet–Naor–Regev, Theorem 1.1, proves that if $p_1, p_2, p_3 \in (1, \infty)$ and

$$\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} \leq 1,$$

then

$$\ell_{p_1} \widehat{\otimes} \ell_{p_2} \widehat{\otimes} \ell_{p_3}$$

does not have finite cotype. In particular, with $p_1 = p_2 = p_3 = 3$, the space

$$B = \ell_3 \widehat{\otimes} \ell_3 \widehat{\otimes} \ell_3$$

does not have finite cotype.

Identification

The theorem above contradicts the guessed triple Rademacher formula in the natural type-2 UMD class.

Indeed, ℓ_3 has finite cotype, so the Rademacher and Gaussian sequence spaces $R(\ell_3)$ and $G(\ell_3)$ coincide up to equivalent norms. If Pisier's guessed formula held for $B = \ell_3 \widehat{\otimes} \ell_3 \widehat{\otimes} \ell_3$, then every sequence in $R(B)$ would be a sum of three projective-tensor terms, for example a term in $R(\ell_3) \widehat{\otimes} \ell_3 \widehat{\otimes} \ell_3$, and similarly for the other two coordinates. Replacing $R(\ell_3)$ by $G(\ell_3)$ and using the projective tensor norm shows that each such term is Gaussian-convergent in B . Hence $R(B) \subseteq G(B)$.

The reverse inclusion $G(B) \subseteq R(B)$ holds in every Banach space. Thus the guessed formula would imply $R(B) = G(B)$. By the Maurey–Pisier criterion quoted in the source paper, this is equivalent to finite cotype of B . This contradicts Briet–Naor–Regev's Theorem 1.1 for $p_1 = p_2 = p_3 = 3$. Therefore the proposed triple formula cannot hold in general, even for the classical choice $E = F = G = \ell_3$.

Scope

This is not a new theorem; it is a direct implication from later literature. The supporting paper explicitly studies finite-cotype questions for projective tensor products and cites the random-series source as one place where such questions were stated, but it does not phrase its theorem as a disproof of the triple $R(E \widehat{\otimes} F \widehat{\otimes} G)$ formula.

This packet does not answer the Hilbert triple problem $\ell_2 \widehat{\otimes} \ell_2 \widehat{\otimes} \ell_2$, the two-fold sub-2 question for $L_p \widehat{\otimes} L_q$, or the separate second-derivative formula for $R_1(R_2(E \widehat{\otimes} F))$.

References

1. G. Pisier, *Random Series of Trace Class Operators*, arXiv:1103.2090.
2. J. Briet, A. Naor, and O. Regev, *Locally decodable codes and the failure of cotype for projective tensor products*, arXiv:1208.0539.